Count-Only Inverse-Variance Weighting for Intervention Harvesting

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Abstract

Intervention-harvesting estimators of position bias [1] combine per-cell click-rate estimates across many $(q, d, \text{position})$ cells whose impression counts span several orders of magnitude. Under the Position-Based Model (PBM), with cell-level relevance treated as homoscedastic, the *harmonic* weight $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ is the count-only inverse-variance choice—Aitken's classical generalised-least-squares solution [2] restricted to count-only weights—and an unconditional Cauchy–Schwarz inequality shows it minimises the delta-method variance *surrogate* $V_k + V_{k'}$ for any impression counts. We propose the plug-in delta-method asymptotic variance $\widetilde{V}(\omega)$ of the self-normalised weighted ratio estimator [9, 11] as a finite-sample diagnostic that is well aligned with the conditional-on-counts efficiency question, in contrast to a naive pair-bootstrap. On the full KDD Cup 2012 Track 2 sponsored-search log (235M impressions, 4.6M interventional pairs), the plug-in asymptotic relative efficiency of harmonic over uniform is 4.53, well-calibrated against click-resample standard error (1–2% at two sample fractions), and robust to weight-capping (ARE $\geq$ 4.1 at the 99.99th percentile). A Cochran-Q diagnostic confirms substantial PBM mis-specification ($Q/\text{df} = 3.83$), allowing efficiency comparisons to be interpreted separately from model fit; under such mis-specification different weight schemes target different weighted averages of cell-specific ratios—an estimand caveat we discuss explicitly. Synthetic (43%–54% MSE reduction at high imbalance) and Yahoo-LTR semi-synthetic ($\sim$ 40% variance reduction at $r = 50$) experiments with ground truth corroborate the efficiency prediction.

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1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [6]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions $(1, 2)$), **60.7% of pairs have $N_1 + N_2 \leq 5$ while 0.1% of pairs carry 42% of all impressions** [7]. Public benchmarks hide this shape: the released Baidu-ULTR [14] aggregation has 94% of (q, d, pos) cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$ *because* the publication pipeline deduplicates impressions per cell—an artefact of release engineering, not of the underlying production logs. The unaggregated KDD release preserves the heavy-tailed production distribution.

Equal-weighting heteroscedastic estimators is statistically sub-optimal by a century of classical theory: Aitken's generalised-least-squares theorem [2] mandates inverse-variance weights, a result the meta-analysis literature [3, 5] has championed for fifty years. Recent ULTR work [1, 12] introduced *harmonic* weighting $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ to reduce variance versus uniform on heavy-tailed counts, but existing validation has relied on *bootstrap* resampling, which we show can be misleading for inverse-variance-style weights on singleton-dominated heavy-tailed data (Section 3).

Contributions. (1) We identify harmonic as the count-only inverse-variance choice under the standard PBM with homoscedastic relevance (Theorem 2), prove an unconditional Cauchy–Schwarz reduction of the delta-method variance *surrogate* $V_k + V_{k'}$ (Theorem 3, conditional on the surrogate objective and unconditional in the counts), and characterise the exact asymmetric surrogate oracle of which harmonic is the $\alpha = \beta$ special case (Theorem 4). **(2)** We adopt the plug-in delta-method asymptotic variance $\widetilde{V}(\omega)$ as a finite-sample efficiency diagnostic aligned with the conditional-on-counts question, and use empirical click-resample SE as its calibration baseline; we discuss why pair-bootstrap is misaligned with this question on heavy-tailed data. **(3)** On the full KDD log we measure $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$, calibrated within 1–2% across two sample fractions, and robust to weight-capping at the 99.99th percentile (ARE $\geq$ 4.1). We quantify PBM mis-specification via Cochran's $Q/\text{df} = 3.83$ and discuss the estimand caveat that follows. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Under PBM, $P(\text{click} \mid q, d, k) = p_k\, r_{q,d}$. Given an interventional set $I_{k,k'}$ of (q, d) pairs that appeared at both positions k and k', the position-bias ratio is estimated as [1]

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i\, \hat{\theta}_k^i}{\sum_i \omega_i\, \hat{\theta}_{k'}^i}, \quad \hat{\theta}_k^i = \frac{C_k^i}{N_k^i}. \tag{1}$$

This is a self-normalised weighted ratio estimator. We say a weight scheme is *ratio-unbiased* when the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$, i.e. $\mathbb{E}[A \mid \mathbf{N}]/\mathbb{E}[B \mid \mathbf{N}] = p_k/p_{k'}$ (*not* $\mathbb{E}[\hat{R}] = p_k/p_{k'}$; the finite-sample bias of the self-normalised ratio is $O(1/N)$ under standard regularity). Under PBM, the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$ for any count-only weight $\omega_i = \omega(N_k^i, N_{k'}^i)$, and

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the resulting self-normalised ratio estimator is consistent for this target [1].

Proposition 1 (Conditional delta-method variance). *By the delta method [11],* $\mathrm{Var}(\hat{R} \mid \mathbf{N}) \approx R^2[\alpha V_k(\omega) + \beta V_{k'}(\omega)]$ *with* $V_r(\omega) = \sum_i \omega_i^2/N_r^i/(\sum_i \omega_i)^2$, $\alpha = (1-p_k)/p_k$, $\beta = (1-p_{k'})/p_{k'}$.

Harmonic as count-only IVW.. For Bernoulli observations, $\mathrm{Var}(\hat{\theta}_k^i) = \theta_k^i(1-\theta_k^i)/N_k^i$. Under PBM, $\theta_k^i = p_k r_i$, so the Bernoulli variance contains a cell-level relevance r_i that is generally heterogeneous. After suppressing this cell-level relevance heterogeneity, the inverse-variance optimum of Aitken [2] restricted to count-only weights is the harmonic mean of the two cell counts. We make this precise as a statement about the variance *surrogate* obtained by ignoring cell-level relevance variation:

Theorem 2 (Count-only IVW on the surrogate). *Among all non-negative weights* $\omega_i = \omega(N_k^i, N_{k'}^i)$, *harmonic weighting minimises the count-only delta-method variance surrogate* $V_k(\omega)+V_{k'}(\omega)$.

Theorem 3 (Surrogate guarantee, unconditional in counts). *For any impression counts* $\{N_k^i, N_{k'}^i\}_{i=1}^n$, $V_k(\omega^h)+V_{k'}(\omega^h) \le V_k(\mathbf{1})+V_{k'}(\mathbf{1})$. *The guarantee is unconditional in the counts but conditional on* $V_k + V_{k'}$ *as the objective; it implies improved* $\mathrm{Var}(\hat{R} \mid \mathbf{N})$ *under PBM when* $\alpha \approx \beta$ *(Proposition 1 below).*

Sketch. Substituting $\omega_i^h = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$ collapses the LHS to $1/W$ with $W = \sum_i \omega_i^h$. The RHS is $(1/n^2)\sum_i 1/\omega_i^h$. Cauchy–Schwarz gives $n^2 \le W \sum_i 1/\omega_i^h$. □

Theorem 4 (Exact asymmetric oracle). $\alpha V_k(\omega) + \beta V_{k'}(\omega) \ge (\sum_i 1/s_i)^{-1}$ *with* $s_i = \alpha/N_k^i + \beta/N_{k'}^i$, *attained at* $\omega_i^\star \propto N_k^i N_{k'}^i/(\alpha N_{k'}^i + \beta N_k^i)$. *Harmonic is the* $\alpha = \beta$ *special case.*

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count heuristic of [1], which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true α, β into $\omega^\star$ yields variance within ±2% of harmonic. Complete proofs and additional sensitivity analyses are provided in the supplementary material.

3 Plug-in Asymptotic Variance: the Right Instrument

$\widetilde{V}$ *in closed form.* With $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at k and k' conditionally independent given counts, the delta method gives

$$\widetilde{V}(\omega) = \frac{\sum_i \omega_i^2\, \theta_k^i(1-\theta_k^i)/N_k^i}{A^2} + \frac{\sum_i \omega_i^2\, \theta_{k'}^i(1-\theta_{k'}^i)/N_{k'}^i}{B^2}, \quad (2)$$

$\mathrm{SE}(\hat{R}) = |\hat{R}|\sqrt{\widetilde{V}(\omega)}$. Plugging in $\hat{\theta}_i$ (with light shrinkage toward the marginal) yields the textbook self-normalised IS variance estimator [8–10].

ARE.. For schemes a, b, $\mathrm{ARE}(a,b) = \widetilde{V}(\omega_b)/\widetilde{V}(\omega_a)$: scheme b needs $\mathrm{ARE}(a,b)$ times the data of a for equal precision. $\widetilde{V}$ scales as $1/\mathrm{ESS}(\omega)$, with Kong's effective sample size [8] $\mathrm{ESS}(\omega) = (\sum_i \omega_i)^2/\sum_i \omega_i^2$.

Table 1: Plug-in delta-method asymptotic variance $\widetilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\text{pairs}} = 4.61$M). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5. The full-θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min [1]	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	**4.53**	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ-aware oracle	2.343	0.148	0.002	1.1

Table 2: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ tertile on full KDD (the bottom quartile boundary collapses into the lowest tertile because singletons dominate). Schemes are tied on the noise-dominated strata; harmonic dominates $\sim 5\times$ on the high-information stratum where the actual estimation signal lives.

$N_1 + N_2$ tertile	n pairs	$\widetilde{V}_U$	$\mathrm{ARE}(h,u)$
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	**4.98**

Why pair-bootstrap is misaligned here. A naive bootstrap that resamples (q,d) pairs estimates a different quantity than the conditional-on-counts efficiency we care about, for three reasons. *(i)* Averaging over $\sim 60\%$ near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects averaging-of-noise rather than estimator precision. *(ii)* Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. *(iii)* Pair-bootstrap entangles estimator efficiency with model fit; $\widetilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample (parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

Calibration. Empirical click-resample SE and the plug-in $\widetilde{V}$ agree closely on KDD: at $f = 0.001$ within 1% (uniform 0.1306 vs. 0.1321; harmonic 0.1113 vs. 0.1120); at $f = 0.01$ within 2% (uniform 0.0387 vs. 0.0379; harmonic 0.0285 vs. 0.0280). The plug-in is well calibrated in this diagnostic.

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \text{ad}, \text{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair (1, 2) interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \le 5$.

Headline (Table 1). $\mathrm{ARE}(\text{harmonic}, \text{uniform}) = 4.53$. **Uniform requires** $\sim 4.5\times$ **the impressions of harmonic for matching precision**, with $\mathrm{SE}_U/\mathrm{SE}_H \approx 2.13$. The min heuristic [1] achieves

4.41, the count-only oracle 4.57; harmonic is the principled parameter-free optimum.

Stratified ARE (Table 2). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by 5× on the high-information stratum. The gain comes from *not* treating singletons as equal to million-impression cells.

Weight-cap robustness. The harmonic ESS of 1,849 out of 4.61M pairs invites a concern that the efficiency gain rests on a few extreme cells, but the small ESS reflects deliberate concentration on the pairs containing nearly all statistical information in this heavy-tailed setting rather than instability of the estimator. Capping ω_i^h at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to 5.2×10^4 and still gives ARE = 4.12; at the 99th percentile, ARE = 2.83 (ESS 7.6×10^5); even at the 90th, ARE = 1.86. Harmonic beats uniform robustly across capping levels.

Estimand under PBM mis-specification. PBM is a known approximation; on real logs different weight schemes estimate different functionals of cell-specific ratios. Specifically, the population limit of $\hat{R}$ under any weight ω is

$$R_\omega = \frac{\sum_i \omega_i \, \theta_k^i}{\sum_i \omega_i \, \theta_{k'}^i},$$

which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is the same constant across cells—i.e. when PBM holds exactly. Under mis-specification each scheme thus targets a different weighted average of the cell-specific ratios $\rho_i = \theta_k^i/\theta_{k'}^i$ (uniform → 1.68, harmonic → 1.71, θ-aware oracle → 2.34 on KDD). The plug-in $\widetilde{V}$ comparison is a precision statement at fixed ω—a model-based efficiency comparison—and does *not* claim that one R_ω is the "correct" position-bias ratio. Choosing ω thus involves a precision/estimand trade-off; harmonic's attractive property is that, among count-only schemes, it concentrates on high-information cells (where ρ_i is estimated most precisely) without depending on noisy plug-in $\hat{\theta}_i$.

Quantifying PBM mis-specification (Cochran's Q). PBM is a known approximation, not a literal model – prior work [4, 12, 13] has long established that real logs violate it. The relevant question is how *large* the deviation is. Cochran's Q on the 16,132 high-information pairs (≥ 5 clicks at each of positions 1 and 2) gives $Q = 6.17\times10^4$, df = 16,131, $Q/\text{df} = 3.83$: per-cell odds-ratio variability is ∼ 3.8× what binomial sampling alone would produce, well outside Poisson dispersion. This magnitude is the quantitative driver of the estimand drift discussed in Paragraph 4; plug-in $\widetilde{V}$ keeps the efficiency comparison well-posed by working at fixed ω.

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1], M=10 positions, $p_k = k^{-1}$, anchor-item model with traffic split s and imbalance ratio r, 500 trials. At $s = 0.80$, $r = 100$ harmonic reduces variance by 43.6% vs. uniform, 5% vs. min; reduction grows to ∼ 54% at $s = 0.90$, $r = 50$. Across the (s, r) grid, the data-equivalence factor DEF = $b_{\text{uni}}/b_{\text{harm}}$ has median 1.31 and maximum 1.94.

Yahoo-LTR semi-synthetic. Same simulator but with real graded relevance from Yahoo Learning to Rank Challenge Set 1 (6936 test queries; clicks simulated via PBM). Harmonic reduces variance by ∼ 40% vs. uniform at imbalance $r = 50$ (Var 0.00015 → 0.00009; ∼458× lower MSE than aggressive clipping). Production-style traffic simulation with K=5 concurrent rankers (one default at 80% + four treatment at 5% each, swap probability 0.5): under uniform traffic only 0.4% of pairs exceed ratio 10, but under production traffic 16.7% do; harmonic's MSE reduction grows from 6.3% to **17.2%**, confirming the gain scales with traffic-induced imbalance.

6 Conclusion

Framing harmonic weighting as the count-only inverse-variance optimum (Theorems 2–4) and replacing bootstrap with the plug-in delta-method variance $\widetilde{V}$ (Eq. 2) yields rigorous, calibration-verified efficiency comparisons on heavy-tailed real click logs. On the full KDD Cup 2012 Track 2 log, ARE(harmonic, uniform) = 4.53, rising to ∼ 5× on information-bearing cells, with PBM mis-specification quantified separately by Cochran-Q ($Q/\text{df} = 3.83$). Future work: shrinkage variants of the θ-aware oracle (Theorem 4 with $\hat{\theta}_i$ regularised); click models beyond PBM [4, 12, 13] where the intervention-harvesting family must be reformulated.

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code, scripts, and data-processing pipelines are provided as supplementary material.

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